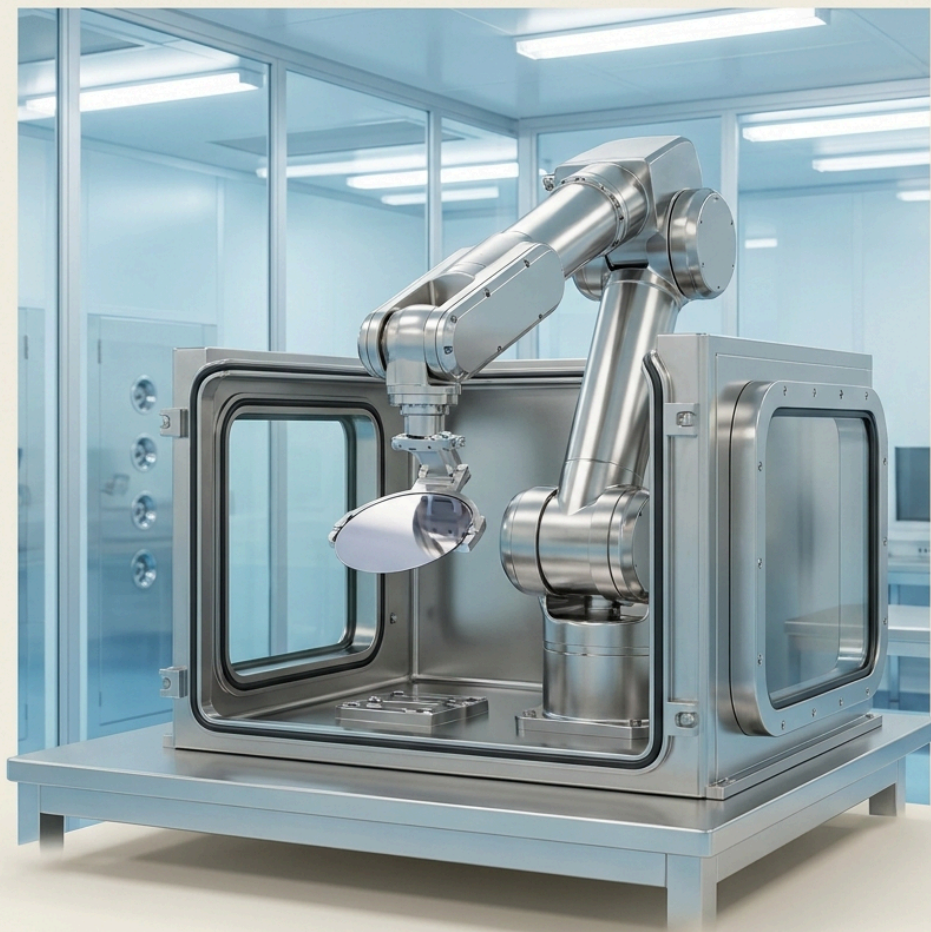


LEDGER · 2026-09-01 · SPECIALTY CHEMICALS / TRIBOLOGY

Halogen-Free Orthoborate-Phosphonium Ionic Liquid ([P6,6,6,14][BMB]) via Ambient Aqueous Metathesis

CLEARED



This concept cleared all seven gates and every voiding sub-test.

Incumbent benchmark

Chemours Krytox GPL 106

Entry application

Boundary lubrication for high-vacuum semiconductor manufacturing robotics and aerospace actuators

Measured advantage	>10x reduction in ball wear volume under 150 N extreme load (wear scar diameter approximately equals theoretical Hertzian contact diameter compared to significant wear for Krytox GPL 106)
Time to first revenue	6 months
Gross margin at parity	88.4%
Startup CapEx	\$103,500
Payback from first sale	0.3 months
Capital productivity	56.52x
Regulatory risk	Moderate — qualification costs reported (\$45,000 estimated)

BUY THE FULL DOSSIER

Mechanism, operating protocol with real conditions and yields, computed unit economics with a six-scenario stress test, the absence audit, and every resolved citation.

[Buy the full dossier — \\$999.00 →](#)

Assessed 2026-09-01 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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